

Selenium

Fact Sheet for Health Professionals

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This is a fact sheet intended for health professionals. For a general overview, see our [consumer fact sheet](#).

Introduction

Selenium is an essential mineral that is naturally present in many foods and added to others. It is also available as a dietary supplement. Selenium is a constituent of 25 selenoproteins, including thioredoxin reductases, glutathione peroxidases, and selenoprotein P [1]. Selenoproteins play critical roles in thyroid hormone metabolism, DNA synthesis, reproduction, and protection from oxidative damage and infection [2-4].

Soil and ground water contain inorganic forms of selenium (e.g., selenites, selenates) that plants accumulate and convert to organic forms, mostly selenomethionine and selenocysteine and their methylated derivatives. In foods, selenium is present primarily as selenomethionine along with selenocysteine. Dietary selenium is readily absorbed by the body, and absorption is largely not affected by selenium status [4-6].

Assessing selenium status

Plasma and serum selenium concentrations are commonly used to assess selenium status, and concentrations of 8 mcg/dL or higher are usually considered sufficient in healthy people. Hair and nail selenium content can be used to assess long-term selenium intakes over months or years. Two plasma selenoproteins,

glutathione peroxidase 3 and selenoprotein P, can be used as functional biomarkers of selenium status, but they can be affected by factors such as inflammation, making them potentially unreliable.

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Recommended Intakes

The Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine has established Recommended Dietary Allowances and Adequate Intakes for selenium. These values range from 55 to 70 mcg for adults and from 15 to 70 mcg for infants, children, and adolescents, depending on age and life stage.

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Sources of Selenium

Food

Foods that are high in protein, such as seafood, meat, and poultry, are rich sources of selenium. Brazil nuts contain particularly high amounts of selenium. The selenium concentrations of plant-based foods vary by geographic location because the amount and form of selenium in the soil and other factors can influence the selenium content of these foods.

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Dietary Supplements

Forms of selenium that are commonly found in dietary supplements include selenomethionine, selenium-enriched yeast, sodium selenite, and sodium selenate. The body is able to absorb up to about 90% of the selenium from these forms.

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Selenium Intakes and Status

While selenium intakes and serum concentrations can vary by region, most people in the United States consume adequate amounts of selenium. However, people who follow a vegetarian or vegan dietary pattern may have lower selenium intakes, and people who smoke tend to have lower selenium status than those who do not, possibly because smoking increases oxidative stress.

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Selenium Deficiency

On its own, selenium deficiency rarely causes illness. However, in certain populations, selenium deficiency may trigger the development of conditions such as Keshan disease and Kashin-Beck disease. In addition, selenium deficiency may increase the risk of congenital hypothyroidism in infants by exacerbating iodine deficiency.

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Groups at Risk of Selenium Inadequacy

Although selenium deficiency is uncommon in the United States, some groups of people have a higher risk of inadequate intakes. These groups include people who live in selenium-deficient regions, those undergoing kidney dialysis, and those living with HIV.

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Selenium and Health

This section focuses on six diseases and conditions in which selenium might play a role: [cancer](#), [cardiovascular disease \(CVD\)](#), [cognitive decline and Alzheimer's disease \(AD\)](#), [HIV infection](#), [male fertility](#), and [thyroid disease](#).

Cancer

Some epidemiological studies have suggested that people with higher selenium status have a lower risk of several types of cancer, and one randomized controlled trial reported that men who took selenium had a lower risk of prostate cancer over a multi-year follow-up period. However, subsequent randomized controlled trials

have found that selenium supplementation does not reduce the risk of prostate cancer or other forms of cancer. More research is needed to characterize the relationship between selenium supplementation and cancer risk.

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Cardiovascular disease

The observational studies that have examined the associations between low selenium status and the risk of cardiovascular disease have produced conflicting results. The available clinical trial evidence does not support using selenium supplementation to reduce the risk of cardiovascular disease, especially in people who obtain sufficient selenium from their diet. However, taking selenium as part of an antioxidant formula may reduce the risk of cardiovascular mortality.

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Cognitive decline and Alzheimer's disease

Researchers are investigating the role of selenium in maintaining cognitive function in older adults, because chronic selenium deficiencies have been linked to cognitive decline. The results from observational studies have been mixed, with some reporting an association between lower plasma selenium concentrations and neurological impairments or a higher risk of Alzheimer's disease and others finding no such association. The evidence from clinical trials does not support the use of selenium supplementation to reduce the risk of cognitive decline and dementia in older adults.

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HIV infection

Although selenium deficiency can increase the risk of morbidity and mortality among people with HIV, selenium supplementation appears to provide limited benefits to this population. Clinical trials in people with HIV have reported that selenium supplementation has little, if any, effect on immune function endpoints, and two Cochrane Reviews concluded that there was insufficient evidence to determine whether selenium supplementation is beneficial in people with HIV.

However, some evidence suggests that selenium supplementation may reduce the risk of preterm birth in pregnant women with HIV.

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Male fertility

While some clinical trials have reported improvements in certain aspects of sperm quality among men who received selenium supplements, others have reported no effects. There is currently not enough data to determine whether selenium supplements significantly affect male fertility.

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Thyroid disease

Selenium plays important roles in the synthesis and metabolism of thyroid hormones. In women, low selenium status appears to be associated with an increased risk of thyroid disease; this association has not been found in men. Clinical trials have found that selenium supplementation can lower the levels of certain thyroid antibodies in people with chronic autoimmune thyroiditis, but other trials have reported no effect on thyroid function or the quality of life in these patients. The results of the clinical trials that have examined the effects of selenium supplementation on thyroid antibodies in pregnant women have also been mixed, and the American Thyroid Association recommends against selenium supplementation in pregnant women who have thyroid autoimmunity.

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Health Risks from Excessive Selenium

Selenosis is the result of chronically high intakes of selenium. It is most commonly characterized by hair loss and nail brittleness or loss, but other signs and symptoms can include a garlic odor in the breath, a metallic taste in the mouth, skin rash, nausea, diarrhea, fatigue, irritability, and nervous system abnormalities. The Tolerable Upper Intake Level for selenium is 400 mcg for adults, and it ranges from 45 mcg to 400 mcg for infants, children, and adolescents, depending on age.

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Interactions with Medications

Selenium supplements may interact with medications. In addition, certain medications, such as cisplatin, can affect selenium levels in the body.

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Selenium and Healthful Diets

In general, a person's nutritional needs should be met primarily through the diet, including fortified foods. Dietary supplements may be useful in cases where it is not possible to meet the needs for specific nutrients through food alone, especially during certain life stages. The *Dietary Guidelines for Americans* offers a general description of healthy dietary patterns.

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Budget

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